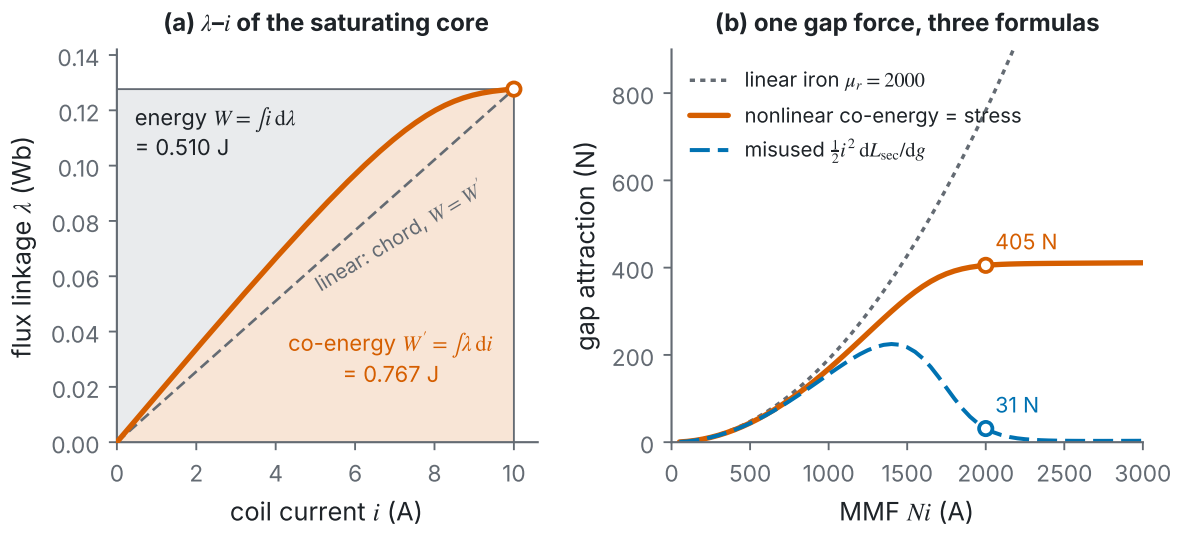
(a) λ–i of the saturating core
flux linkage λ (Wb)
coil current i (A)
energy W = ∫i dλ
= 0.510 J
linear: chord, W = W′
co-energy W′ = ∫λ di
= 0.767 J
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
0
2
4
6
8
10
(b) one gap force, three formulas
linear iron μr = 2000
nonlinear co-energy = stress
misused ½i² dLsec/dg
gap attraction (N)
MMF Ni (A)
405 N
31 N
800
600
400
200
0
500
1000
1500
2000
2500
3000